CompTIA Security + 701



Course-End Project



Implementing SSH Brute-Force Simulation, Log Analysis, and Defensive Hardening on Windows

Objective



Conduct a comprehensive security assessment and incident response for a Windows Server system running OpenSSH during a brute-force attack, focusing on analyzing Windows Security logs, verifying user accounts, and implementing SSH hardening measures.

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Prerequisites



- Windows Server administration
- SSH fundamentals and public key authentication
- Incident response
- Windows security log monitoring (Event Viewer)
- Network security and firewall configuration

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Problem Statement and Scope



Problem Statement:
Conduct a comprehensive security assessment and incident response for a Windows Server system running OpenSSH during a brute-force attack, focusing on analyzing Windows Security logs, verifying user accounts, and implementing SSH hardening measures.

Real-World Scenario:
Your organization’s critical Windows Server has been targeted by an SSH brute-force attack, prompting an urgent need to analyze Windows Security authentication logs. The IT security team will identify and check attempted usernames against the company’s user database to determine whether any valid accounts were targeted.

To counter the attack, the team will implement security enhancements, including certificate-based SSH authentication, limiting authentication attempts, restricting SSH access using Windows Defender Firewall, and performing log-based monitoring using Event Viewer.

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Industry Relevance



This project applies the following industry-relevant skills:

1. **Log analysis and scripting tools:** Use of Windows Security event logs and PowerShell (Get-WinEvent) for detecting brute-force attempts and validating authentication activity using Event Viewer

2. **Network security and access control tools:** Configuration of SSH services and firewall access control using OpenSSH and Windows Defender Firewall to restrict unauthorized access

3. **Certificate-based authentication and user management:** Implementation of SSH public key authentication to eliminate password-based login and strengthen remote access security

4. **Continuous log monitoring and documentation:** Manual monitoring of authentication events through Windows Security logs and documentation of findings in SOC-style reporting format

Tasks



1. Verify network connectivity between the attacker and victim systems
2. Install and configure the OpenSSH server on the Windows target system
3. Create test user accounts and verify baseline SSH authentication
4. Simulate an SSH brute-force attack from Kali Linux
5. Investigate SSH authentication failures using Windows Security logs
6. Create and validate successful authentication events for compromise assessment
7. Implement SSH hardening and access control restrictions
8. Implement and validate certificate-based SSH authentication

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Submission Process



You will have to submit the project in one week.

It is recommended to work in the integrated labs, as they have all the required tools available.

The project can be submitted from the assessment tab by clicking on the **Submit** button.

Provide the document(s) mentioned below:

• Screenshots of the outputs

Thank you